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26 August 2025

Due Thursday 4 September 2025, 11 AM

Linear Algebra, Math 2101-002
Homework set #1

1. (2.5 pts.)

Explain why the set $H^{n \times n} = \{n \times n \text{ symmetric matrices} \}$ is a subspace of $\mathbb{R}^{n \times n}$.

2. (a) (5 pts.)

Let A be a matrix in $\mathbb{R}^{n \times n}$. Show that $A + A^T$ is symmetric and $A - A^T$ is skew-symmetric.

(b) Prove that any matrix $A \in \mathbb{R}^{n \times n}$ can be written as $A = H + S$, where H is symmetric, and S is skew-symmetric.

(c) (1 pt. extra credit)

Show that H , S , in part (b) are unique.

3. (2.5 pts.)

Show that the set $\{\mathbf{x} \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}$ is a subspace of $\mathbb{R}^3$.